

Types of RAG

Presented by Omar Ahmed



Introduction



normal RAG systems works using this workflow

- User Question
- Embedding
- Vector Search
- Top-k chunking
- LLM
- Answer

But the problem is retrieved documents may be wrong, or important chunks may be missed, some Questions require multiple Searches

Standard Rag	**Corrective Rag (CRAG)**	**Self RAG**
Adaptive RAG	**Agentic RAG**	**Graph RAG**
Hybrid RAG	**Modular RAG**	**Recursive RAG**
iterative RAG	**Hierarechical RAG**	**Multi Hop Rag**

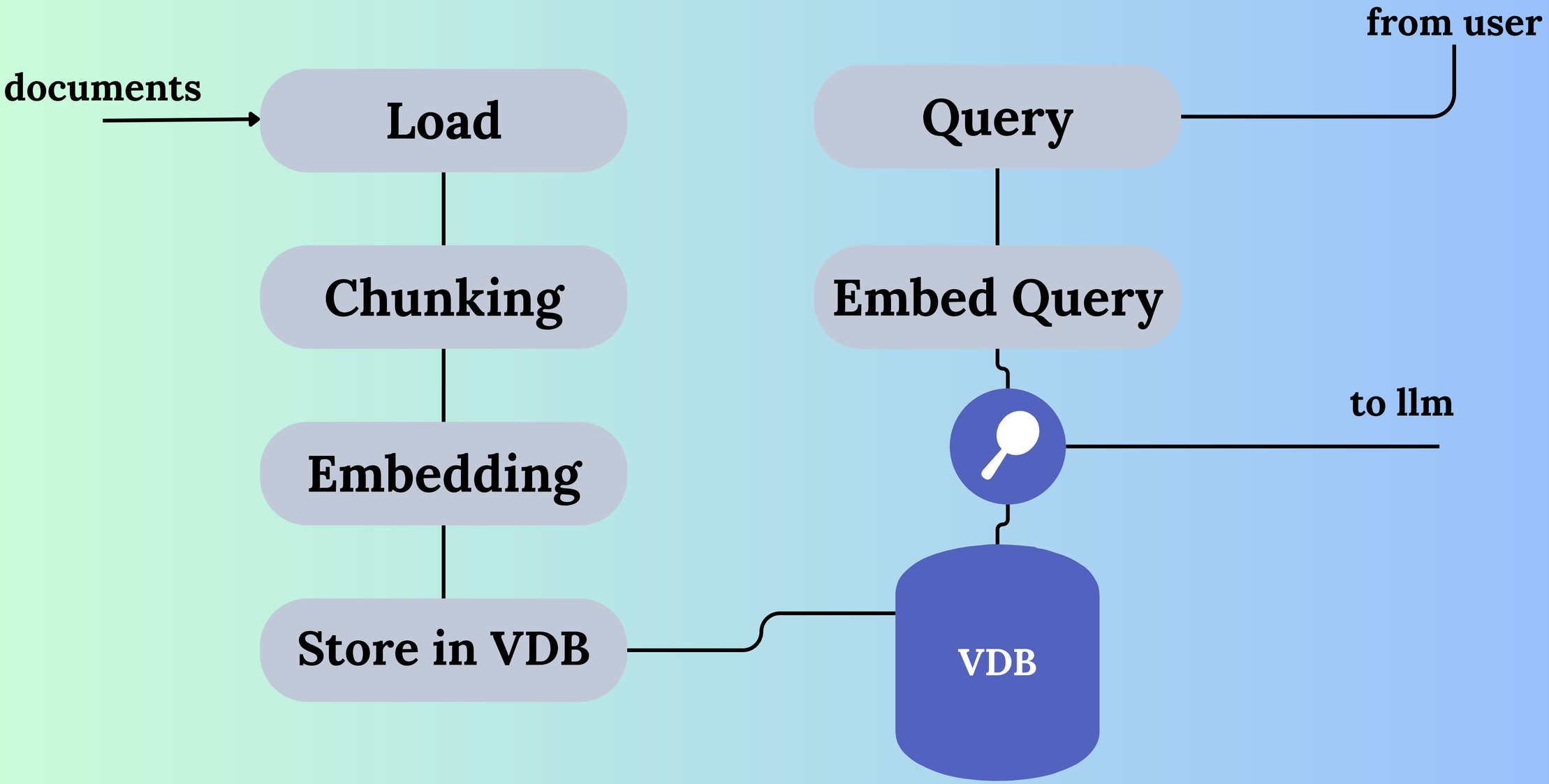
Standard RAG

Online phase

Embed Query
Similariy Search
Prompt construction
Generation

offline phase

Load documents
Chunking
Embedding
Store in Vector DB

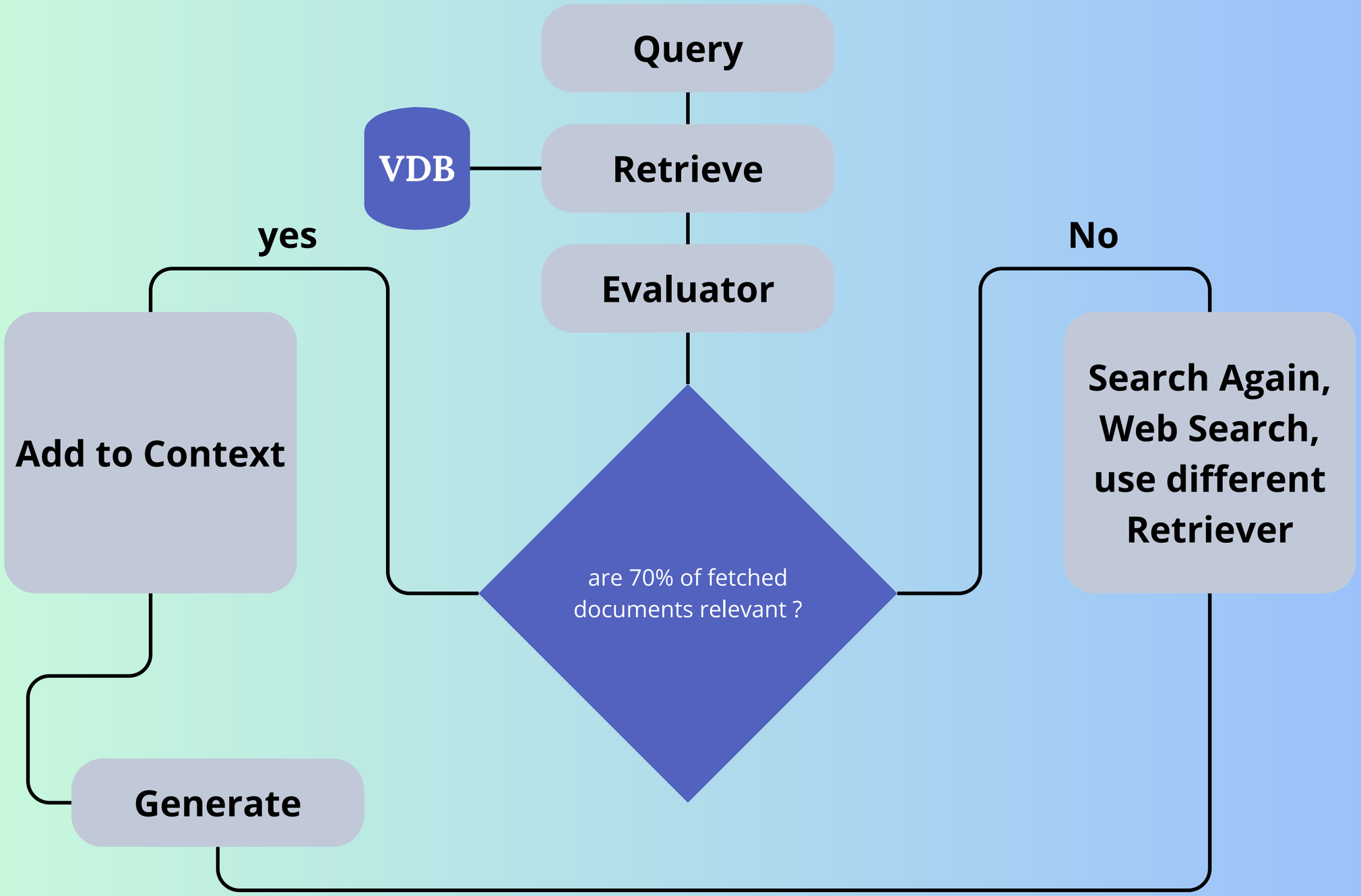

documents
Load
Chunking
Embedding
Store in VDB
from user
Query
Embed Query
to llm
VDB

but what if low similarity between retrived chunks and query ?
llm still answers, and this created hallucinations

so how to fix this ?

what if we add Retriever Evaluator to check similarity scores and based on score we decide if answer is relevant to Query or no ?

this will lead us to CRAG (corrective RAG)


Query
VDB
Retrieve
yes
No
Evaluator
Add to Context
Search Again, Web Search, use different Retriever
are 70% of fetched documents relevant ?
Generate

but wait ?

is it possible to check a huge number of chunks, That would be extremely slow

so production system uses two stages

stage 1 : Fast Retriever to search and retrieves top 50 chunks very quickly

stage 2 : now those 50 chunks are evaluated and sort by score to keep only top k chunks

cross encoder combines Question, Chunk and enter them to Neural Network that output relevance score. so the model reads both text together



Thank You

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